

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
```

```
from sklearn.datasets import load_iris
iris = load_iris()
df = pd.DataFrame(data=iris.data, columns=iris.feature_names)
```

```
print("First 5 rows of dataset:")
print(df.head())
```

```
First 5 rows of dataset:
   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0                5.1                3.5                1.4                0.2
1                4.9                3.0                1.4                0.2
2                4.7                3.2                1.3                0.2
3                4.6                3.1                1.5                0.2
4                5.0                3.6                1.4                0.2
```

```
print("\nDescriptive Statistics:")
print(df.describe())
```

```
Descriptive Statistics:
   sepal length (cm)  sepal width (cm)  petal length (cm)  \
count      150.000000      150.000000      150.000000
mean         5.843333         3.057333         3.758000
std          0.828066         0.435866         1.765298
min          4.300000         2.000000         1.000000
25%          5.100000         2.800000         1.600000
50%          5.800000         3.000000         4.350000
75%          6.400000         3.300000         5.100000
max          7.900000         4.400000         6.900000

   petal width (cm)
count      150.000000
mean         1.199333
std          0.762238
min          0.100000
25%          0.300000
50%          1.300000
75%          1.800000
max          2.500000
```

```
print("\nSkewness of each column:")
print(df.skew())
```

```
Skewness of each column:
sepal length (cm)    0.314911
sepal width (cm)     0.318966
petal length (cm)   -0.274884
petal width (cm)    -0.102967
dtype: float64
```

```
print("\nKurtosis of each column:")
print(df.kurtosis())
```

```
Kurtosis of each column:
sepal length (cm)   -0.552064
sepal width (cm)    0.228249
petal length (cm)  -1.402103
petal width (cm)   -1.340604
dtype: float64
```

```
print("\nCorrelation between variables:")
print(df.corr())
Correlation between variables:
      sepal length (cm)  sepal width (cm)  petal length (cm)  \
sepal length (cm)      1.000000      -0.117570      0.871754
sepal width (cm)       -0.117570      1.000000      -0.428440
petal length (cm)      0.871754      -0.428440      1.000000
petal width (cm)       0.817941      -0.366126      0.962865

      petal width (cm)
sepal length (cm)    0.817941
sepal width (cm)    -0.366126
petal length (cm)    0.962865
petal width (cm)    1.000000
```

```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(), annot=True, cmap='coolwarm')
plt.title("Correlation Heatmap")
plt.show()
```

